

Wordcountprogram.docx

Exported: May 28, 2026

Summary

This technical guide outlines the implementation of a **Hadoop MapReduce application** designed to calculate the frequency of words within a text file. The process is organized into three distinct Java classes: a **Mapper** to tokenize strings into individual units, a **Reducer** to aggregate those units into final counts, and a **Driver** to orchestrate the entire execution. Beyond the programming logic, the document provides essential **HDFS commands** for managing data storage and executing the compiled JAR file within a distributed environment. Ultimately, the source serves as a practical blueprint for performing **distributed data processing** by transforming raw unstructured text into a structured summary of word occurrences.

Keywords: Java Class Creation, MapReduce Programming, Hadoop File System, Word Count Logic, HDFS Command Execution

Content

.Create Three Java Classes into the project. Name them WCDriver(having the main function), WCMapper, WCReducer.// Importing librariesimport java.io.IOException;import org.apache.hadoop.io.IntWritable;import org.apache.hadoop.io.LongWritable;import org.apache.hadoop.io.Text;import org.apache.hadoop.mapred.MapReduceBase;import org.apache.hadoop.mapred.Mapper;import org.apache.hadoop.mapred.OutputCollector;import org.apache.hadoop.mapred.Reporter;public class WCMapper extends MapReduceBase implements Mapper<LongWritable,Text, Text, IntWritable> { // Map function public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter rep) throws IOException { String line = value.toString(); // Splitting the line on spaces for (String word : line.split(" ")) { if (word.length() > 0) { output.collect(new Text(word), new IntWritable(1)); } } }}

Reducer Code: You have to copy paste this program into the WCReducer Java Class

```

file. // Importing librariesimport java.io.IOException;import java.util.Iterator;import
org.apache.hadoop.io.IntWritable;import org.apache.hadoop.io.Text;import
org.apache.hadoop.mapred.MapReduceBase;import
org.apache.hadoop.mapred.OutputCollector;import
org.apache.hadoop.mapred.Reducer;import
org.apache.hadoop.mapred.Reporter;public class WCReducer extends
MapReduceBase implements Reducer<Text,IntWritable, Text, IntWritable> { //
Reduce function public void reduce(Text key, Iterator<IntWritable> value,
OutputCollector<Text, IntWritable> output, Reporter rep) throws IOException { int
count = 0; // Counting the frequency of each words while (value.hasNext())
{ IntWritable i = value.next(); count += i.get(); } output.collect(key, new
IntWritable(count)); }}
```

Driver Code: You have to copy paste this program into the WCDriver Java Class file.

```

// Importing librariesimport java.io.IOException;import
org.apache.hadoop.conf.Configured;import org.apache.hadoop.fs.Path;import
org.apache.hadoop.io.IntWritable;import org.apache.hadoop.io.Text;import
org.apache.hadoop.mapred.FileInputFormat;import
org.apache.hadoop.mapred.FileOutputFormat;import
org.apache.hadoop.mapred.JobClient;import
org.apache.hadoop.mapred.JobConf;import org.apache.hadoop.util.Tool;import
org.apache.hadoop.util.ToolRunner;public class WCDriver extends Configured
implements Tool { public int run(String args[]) throws IOException { if (args.length <
2) { System.out.println("Please give valid inputs"); return -1; } JobConf conf = new
JobConf(WCDriver.class); FileInputFormat.setInputPaths(conf, new Path(args[0]));
FileOutputFormat.setOutputPath(conf, new Path(args[1]));
conf.setMapperClass(WCMapper.class); conf.setReducerClass(WCReducer.class);
conf.setMapOutputKeyClass(Text.class);
conf.setMapOutputValueClass(IntWritable.class); conf.setOutputKeyClass(Text.class);
conf.setOutputValueClass(IntWritable.class); JobClient.runJob(conf); return 0; } //
Main Method public static void main(String args[]) throws Exception { int exitCode =
ToolRunner.run(new WCDriver(), args); System.out.println(exitCode); }}
```

3. create a file on Desktop(sample.txt) and type the below lines:hi how are youhow is your jobhow is your familyhow is your brotherhow is your sistersave your file in /home/hadoop/input

5. View the directory contenthadoop fs -ls /6. Create a directory using the following command. If any directory existing, use the same directory for the command hadoop fs -mkdir /inputbda

7. Copy the file into HDFS hadoop fs -copyFromLocal /input/sample.txt /inputbda orhadoop fs -put /input/sample.txt /inputbdahadoop fs -ls /inputbda

8.Run the Map Reduce Programhadoop jar /home/hduser/Desktop/wordcount.jar WCDriver /inputbda/sample.txt /outputbda

9. hadoop fs -ls /outputbda

10. hadoop fs -cat /outputbda/part-00000